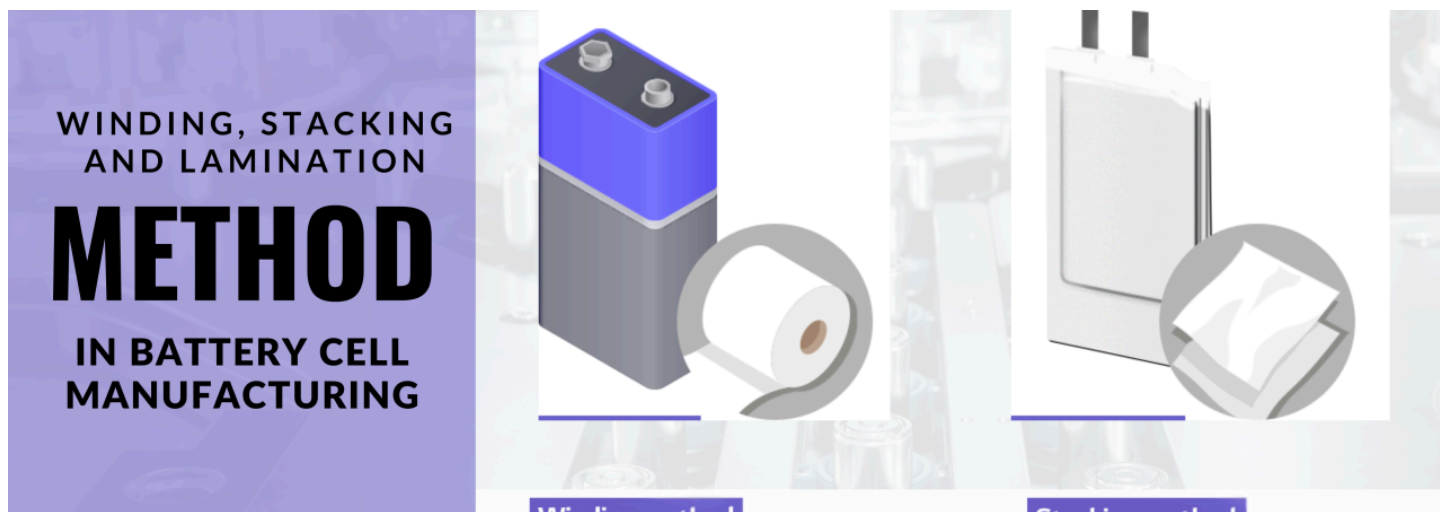




Winding, Stacking and Lamination in Battery Cell Manufacturing



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The global battery industry is experiencing unprecedented growth, with annual battery demand surpassing **1 terawatt-hour** in 2024 and manufacturing capacity reaching **3 TWh**. As we advance into 2025, three critical manufacturing processes—winding, stacking, and lamination—are shaping the future of battery cell production, each offering unique advantages for different applications and performance requirements.

Central to cell assembly are two primary electrode assembly techniques: winding and stacking. Lamination, often closely associated with stacking, refers to the crucial process of bonding these layered materials to create a stable, integrated structure.

Understanding the Three Core Manufacturing Processes

Winding: The "Jelly Roll" Foundation

Winding technology represents the foundational method in lithium-ion battery cell assembly. The process involves rolling positive electrode sheets, negative electrode sheets, and separators together through specialized winding machines equipped with winding needle mechanisms. This creates what's known as a "jelly roll" structure, where the adjacent positive and negative electrode sheets are isolated by separators to prevent short circuits.

The winding process is characterized by several key features:

- **High Efficiency:** Winding offers superior process speed and simplicity compared to alternative methods.

- **Mature Technology:** Cylindrical cells consistently utilize winding technology due to its established reliability.
- **Cost-Effective Production:** The streamlined process enables high-volume manufacturing with lower initial investment.

Modern winding machines incorporate advanced features including active unwinding of electrode materials, automatic tension detection and control, and automated quality inspection systems. The equipment typically achieves efficiency rates of 6 parts per million with qualification rates exceeding 99.5%.

a) Single sheet stacking



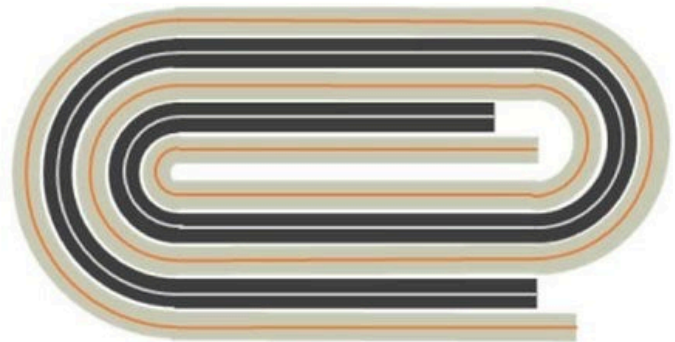
b) Z-stacking



c) Cylindrical winding



d) Prismatic winding



Cathode


Li metal anode


Separator


Stacking Technology: The Precision Alternative

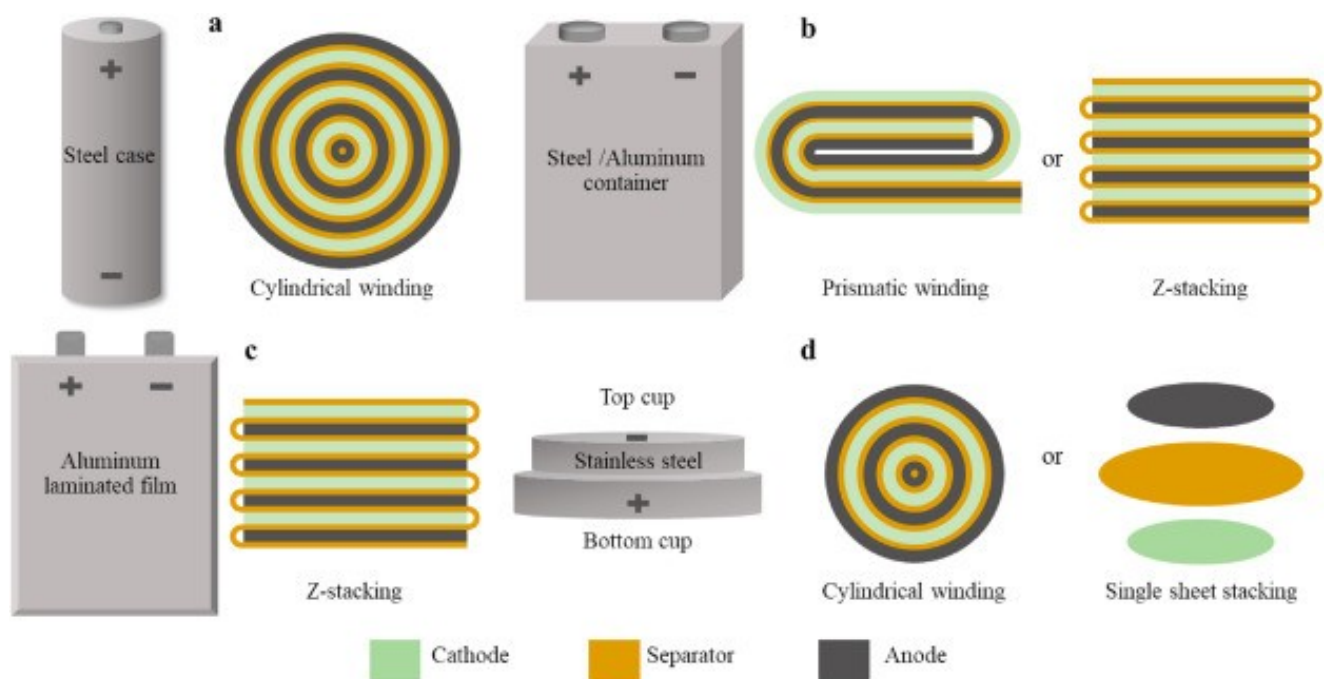
Stacking technology takes a fundamentally different approach by cutting electrode sheets into precise dimensions and stacking them layer by layer with separators in between. This process,

also known as Z-stacking, follows a specific sequence: negative electrode, separator, positive electrode, separator, and so forth.

The stacking process offers several compelling advantages:

- **Superior Energy Density:** Stacking can increase energy density by approximately 5% compared to winding by optimizing space utilization.
- **Enhanced Thermal Management:** The layered structure enables more uniform heat distribution during operation.
- **Lower Internal Resistance:** Multiple tabs in parallel configuration reduce internal resistance by 10-15%.
- **Improved Safety:** The stable internal structure minimizes risks of structural failure and short circuits.

Recent innovations in stacking technology include electromagnetic positioning systems that use eddy currents to precisely position and fix electrodes without contact, significantly reducing process time and preventing damage.



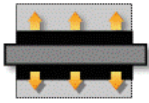
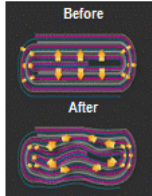
Lamination: The Bonding Process

Lamination serves as a critical complementary process that creates stable, durable structures by layering different materials together under controlled heat and pressure. The lamination process typically operates at temperatures between 80-120°C and pressures of 5-15 MPa.

The lamination process delivers several performance enhancements:

- **Enhanced Electrode Interface Bonding:** Hot pressing eliminates micro-gaps between electrodes and separators, increasing charge/discharge efficiency by 20-30%.

- **Improved Structural Integrity:** The process locks electrodes in optimal alignment, achieving 98.5% first-pass yield rates.
- **Reduced Internal Impedance:** Air-purging mechanisms during lamination lower internal impedance by 12-18%.

	Lamination & Stacking	Winding
Electrode Loading	 Higher electrode loading → Possible to produce high capacity battery	 Electrode loading is limited due to its winding tension
Space Utilization	 No dead space → Higher battery capacity	 When winding, dead space occurs → Lower battery capacity
Electrode Stability	 Dimensional stability after hundreds of cycles → Longer life-cycle	 Electrode distortion occurs after hundreds of cycles → Shorter life cycle

Winding vs. Stacking/Lamination: A Strategic Comparison

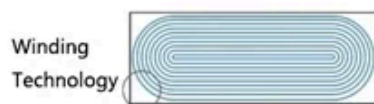
Feature	Winding	Stacking/Lamination
Primary Battery Form Factors	Cylindrical, Prismatic	Pouch, Prismatic, Blade
Energy Density	Lower - Curved corners, less space utilization	Higher (~5-6% more) - Better space utilization, no "C angle"
Internal Resistance	Higher - Longer electron path, typically one tab	Lower - Multiple tabs, parallel connections
Cycle Life	Shorter - Prone to deformation/uneven stress	Longer - Stable internal structure, uniform stress
Safety	Potential for burrs, powder loss, short circuits at "C corners"	Enhanced - Uniform stress, controlled gaps, less bulging
Production Efficiency	High (typically >12 ppm) - Fast	Traditionally Low (6-8 ppm) - Slower <i>*Efficiency improving with new tech</i>
Equipment Cost (Initial Investment)	Lower	Higher (1-3x winding cost) - Larger footprint
Flexibility in Shape	Limited to regular/cylindrical	High - Regular or special shapes
Process Maturity	Very mature	Maturing rapidly - High innovation
Quality Control Difficulty	Simpler cutting - Higher yield	More complex cutting - Burr risk, lower traditional yield

Comparative Analysis: Performance and Applications

Energy Density and Efficiency Comparison

The choice between winding and stacking significantly impacts battery performance characteristics. Stacking technology demonstrates superior performance in several key metrics:

- **Internal Resistance:** Stacking cells exhibit lower internal resistance due to parallel welding of multiple tabs, while winding cells rely on single-tab current output
- **Cycle Life:** Stacking cells show 10-15% longer cycle life compared to wound cells due to more stable electrode structures
- **Rate Capability:** Stacking achieves enhanced rate performance through parallelized current pathways from multiple electrode layers



The Advantages Of Stacking Process

Higher energy density

More stable internal structure



Higher level of safety

Longer life span

Application-Specific Preferences

Different battery form factors favor specific manufacturing approaches:

- **Cylindrical Cells:** Exclusively use winding technology due to geometric requirements and manufacturing maturity
- **Prismatic Cells:** Transitioning from winding to stacking, with Chinese manufacturers leading this shift
- **Pouch Cells:** Employ both technologies depending on manufacturer preference and application requirements
- **Blade Cells:** Specifically designed and produced using stacking processes



Automakers will most likely use prismatic cells



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Cylindrical cell	Prismatic cell	Pouch cell
		
• Small size (e.g. 18650 type (ø 18 mm, height 650 mm)) • Hard casing • Low individual cell capacity • Build in safety features • Comparably cheap	• Hard casing • Large size • High individual cell capacity	• Soft casing • Large size • High individual cell capacity • Geometrical deformation during (dis-)charging

Blade Battery



Equipment and Manufacturing Considerations

Advanced Winding Equipment

Modern winding machines feature sophisticated control systems including AC servo motor drives, tension control unwinding, and real-time monitoring capabilities. Key performance indicators for winding equipment include:

- Winding speed optimization
- Tension fluctuation control (typically $<\pm 3\%$)
- Alignment precision ($<\pm 0.3\text{mm}$)
- Automated quality inspection and defect sorting

Stacking Machinery Innovation

Cell stacking machines represent advanced automation technology that operates through precise feeding, alignment, stacking, compression, and output processes. These machines utilize advanced sensors and robotic arms to ensure accurate cell orientation while minimizing errors and maximizing production speed.

Hot Pressing and Lamination Equipment

Hot pressing equipment enables the synergistic effects of elevated temperature and controlled pressure to enhance battery performance. The process requires precise temperature control systems to ensure uniform heating and prevent diaphragm micropore closure, which could increase internal resistance.



Industry Trends and Future Outlook

The Shift Toward Stacking

Industry analysis reveals a significant trend toward stacking technology, particularly for high-rate applications. Leading lithium battery companies are increasingly adopting stacking methods for large-capacity square, blade, and pouch batteries. The stacking technique has emerged as the second major technological direction in lithium batteries, with penetration rates continuously increasing.

Emerging Manufacturing Trends

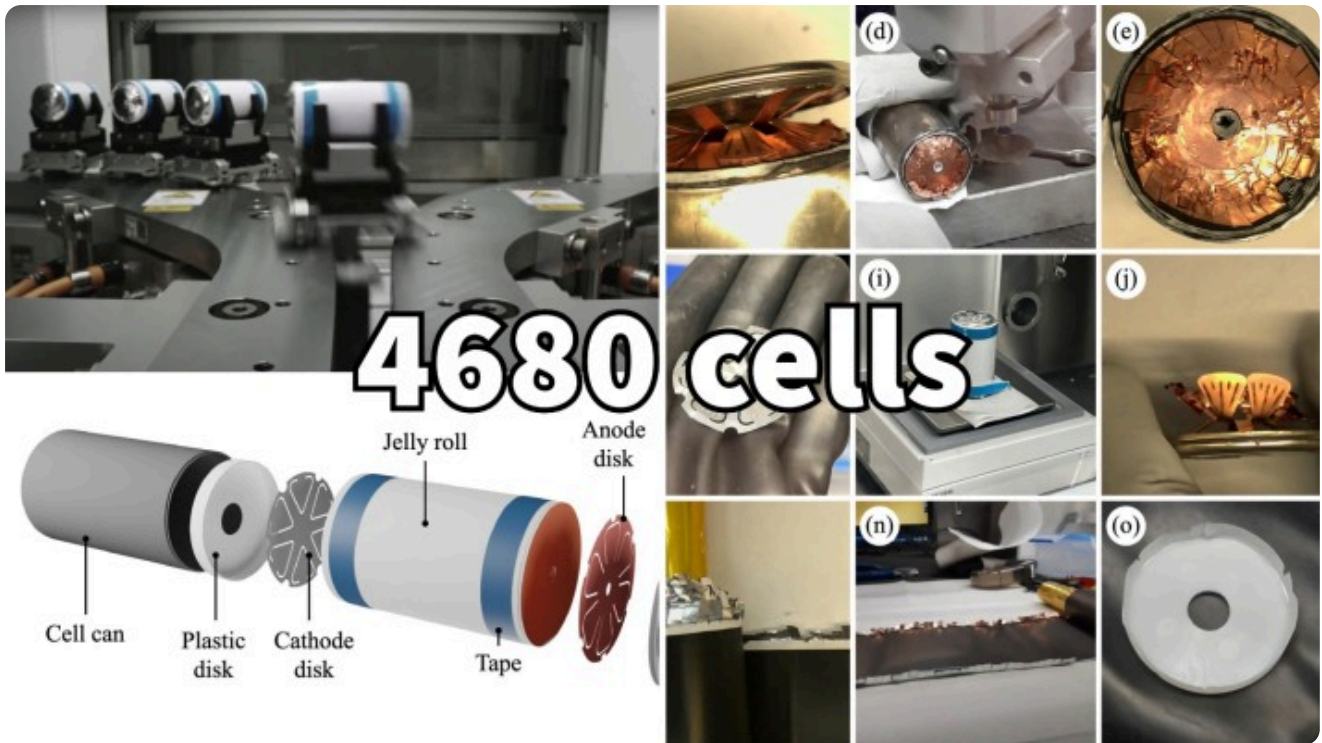
Several key trends are shaping the future of battery manufacturing through 2025:

Dry Electrode Processing: The transition to dry battery electrode (DBE) processing offers reduced energy requirements, smaller factory footprints, and lower production costs. Tesla's adoption of

DBE for 4680 cells represents a significant industry shift, with projected demand reaching 72 GWh by 2025.

Automation and AI Integration: Manufacturing lines increasingly incorporate artificial intelligence and digital twin technologies to optimize production efficiency and quality control.

Sustainable Manufacturing: Growing emphasis on recycling, debond-on-demand technologies, and bio-based materials is reshaping production processes.



Performance Optimization

Recent studies demonstrate that hot-pressed batteries achieve 15% higher discharge capacity under high-rate conditions compared to non-pressed counterparts. This performance improvement, combined with manufacturing efficiency gains, positions lamination as an increasingly critical process step.

Challenges and Opportunities

Manufacturing Challenges

Each technology faces specific challenges:

- **Winding:** Issues with electrode deformation at bending points and uneven current distribution during discharge
- **Stacking:** Lower qualified rates for electrode slitting and higher overall manufacturing costs

- **Lamination:** Precise temperature and pressure control requirements to prevent material degradation

Market Opportunities

The battery industry's rapid expansion creates significant opportunities for manufacturing innovation. Global battery manufacturing capacity could triple over the next five years if all announced projects proceed. This growth, combined with falling battery prices below \$100 per kilowatt-hour, positions advanced manufacturing technologies as critical competitive advantages.

Conclusion: The Integrated Future

The future of battery cell manufacturing lies not in choosing a single technology, but in optimizing the integration of winding, stacking, and lamination processes based on specific application requirements. As the industry moves toward higher energy densities, improved safety, and cost-effective production, manufacturers are developing hybrid approaches that leverage the strengths of each technology.

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